

Investigating the Doppler Effect

SimuPhysics Doppler Effect Simulation · SimuPhysics

Years 8-9

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/oscillations-and-waves-doppler-effect-simulation.html>

Curriculum links: AC9S8U05 · AC9S9U04 · AC9S9U05

Learning intentions

- I can describe how the frequency of a wave changes when its source is moving.
- I can relate the Doppler effect to a wave model of energy transfer.
- I can connect the simulation to real, everyday examples such as a passing siren.

Getting started

Open the simulation and start with the wave source stationary, noting the pattern of wave fronts it produces. Then set the source moving at a slow speed and observe from a fixed observer position.

Tasks

1. With the source stationary, describe the pattern of wave fronts spreading out from it.

2. Set the source moving slowly toward the observer. What changes about the spacing of the wave fronts on the side the source is moving toward?

3. What happens to the wave front spacing on the side the source is moving away from?

4. Using the word 'frequency', explain why an observer in front of the moving source perceives a higher pitch, while an observer behind it perceives a lower pitch.

5. Increase the source speed across three settings and record how closely spaced the wave fronts appear ahead of the source at each speed.

Source speed	Wave front spacing ahead of source

6. Based on your table, what is the relationship between source speed and how much the frequency appears to shift?

7. Explain how the Doppler effect explains why an ambulance siren sounds higher-pitched as it approaches you and lower-pitched as it drives away.

Extension (optional)

The Doppler effect also applies to light, not just sound. Briefly research 'redshift' in astronomy and explain, using what you learned about moving wave sources in this simulation, why a star moving away from Earth appears redder.